2.37 Partial Fractions

Worked Solutions 1–32

1)

$\frac{7x+17}{(x+1)(x+3)} \equiv \frac{A}{x+1}+\frac{B}{x+3}$. $7x+17 = A(x+3)+B(x+1)$. When $x=-3$, $-4=-2B \therefore B=2$. When $x=-1$, $10=2A \therefore A=5$. $\therefore \frac{7x+17}{(x+1)(x+3)} \equiv \frac{5}{x+1}+\frac{2}{x+3}$.

2)

$\frac{12x-3}{(x-2)(x+1)} \equiv \frac{A}{x-2}+\frac{B}{x+1}$. $12x-3 = A(x+1)+B(x-2)$. When $x=-1$, $-15=-3B \therefore B=5$. When $x=2$, $21=3A \therefore A=7$. $\therefore \frac{12x-3}{(x-2)(x+1)} \equiv \frac{7}{x-2}+\frac{5}{x+1}$.

3 a)

$\frac{2x+3}{(x-2)(x+5)} \equiv \frac{A}{x-2}+\frac{B}{x+5}$. $2x+3 = A(x+5)+B(x-2)$. When $x=-5$, $-7=-7B \therefore B=1$. When $x=2$, $7=7A \therefore A=1$. $\therefore \frac{2x+3}{(x-2)(x+5)} \equiv \frac{1}{x-2}+\frac{1}{x+5}$.

3 b)

$\frac{16x+1}{(x+4)(2x+1)} \equiv \frac{A}{x+4}+\frac{B}{2x+1}$. $16x+1 = A(2x+1)+B(x+4)$. When $x=-\frac{1}{2}$, $-7=\frac{7}{2}B \therefore B=-2$. When $x=-4$, $-63=-7A \therefore A=9$. $\therefore \frac{16x+1}{(x+4)(2x+1)} \equiv \frac{9}{x+4}-\frac{2}{2x+1}$.

3 c)

$\frac{4}{(x+2)(x+7)} \equiv \frac{A}{x+2}+\frac{B}{x+7}$. $4 = A(x+7)+B(x+2)$. When $x=-7$, $4=-5B \therefore B=-\frac{4}{5}$. When $x=-2$, $4=5A \therefore A=\frac{4}{5}$. $\therefore \frac{4}{(x+2)(x+7)} \equiv \frac{4}{5(x+2)}-\frac{4}{5(x+7)}$.

3 d)

$\frac{14}{(x+2)(x+6)} \equiv \frac{A}{x+2}+\frac{B}{x+6}$. $14 = A(x+6)+B(x+2)$. When $x=-6$, $14=-4B \therefore B=-\frac{7}{2}$. When $x=-2$, $14=4A \therefore A=\frac{7}{2}$. $\therefore \frac{14}{(x+2)(x+6)} \equiv \frac{7}{2(x+2)}-\frac{7}{2(x+6)}$.

3 e)

$\frac{4}{x(x-2)} \equiv \frac{A}{x}+\frac{B}{x-2}$. $4 = A(x-2)+Bx$. When $x=2$, $4=2B \therefore B=2$. When $x=0$, $4=-2A \therefore A=-2$. $\therefore \frac{4}{x(x-2)} \equiv -\frac{2}{x}+\frac{2}{x-2}$.

3 f)

$\frac{x+9}{x^2+8x+15} = \frac{x+9}{(x+5)(x+3)} \equiv \frac{A}{x+5}+\frac{B}{x+3}$. $x+9 = A(x+3)+B(x+5)$. When $x=-3$, $6=2B \therefore B=3$. When $x=-5$, $4=-2A \therefore A=-2$. $\therefore \frac{x+9}{x^2+8x+15} \equiv -\frac{2}{x+5}+\frac{3}{x+3}$.

3 g)

$\frac{49-x}{2x^2+13x-24} = \frac{49-x}{(2x-3)(x+8)} \equiv \frac{A}{2x-3} + \frac{B}{x+8}$. $49-x = A(x+8)+B(2x-3)$. When $x=-8$, $57 = -19B \therefore B = -3$. When $x = \frac{3}{2}$, $\frac{95}{2} = \frac{19}{2}A \therefore A = 5$. $\therefore \frac{49-x}{2x^2+13x-24} \equiv \frac{5}{2x-3} - \frac{3}{x+8}$.

3 h)

$\frac{2(11x+4)}{8x^2+6x+1} = \frac{22x+8}{(4x+1)(2x+1)} \equiv \frac{A}{4x+1} + \frac{B}{2x+1}$. $22x+8 = A(2x+1)+B(4x+1)$. When $x = -\frac{1}{2}$, $-3 = -B \therefore B = 3$. When $x = -\frac{1}{4}$, $\frac{5}{2} = \frac{1}{2}A \therefore A = 5$. $\therefore \frac{2(11x+4)}{8x^2+6x+1} \equiv \frac{5}{4x+1} + \frac{3}{2x+1}$.

3 i)

$\frac{k}{x^2+x-20} = \frac{k}{(x+5)(x-4)} \equiv \frac{A}{x+5} + \frac{B}{x-4}$. $k = A(x-4)+B(x+5)$. When $x=4$, $k = 9B \therefore B = \frac{k}{9}$. When $x=-5$, $k = -9A \therefore A = -\frac{k}{9}$. $\therefore \frac{k}{x^2+x-20} \equiv -\frac{k}{9(x+5)} + \frac{k}{9(x-4)}$.

4 a)

In his second line of working he incorrectly rearranged, putting $25-4x^2$ rather than 1 on the left hand side of the equation.

4 b)

$\frac{1}{25-4x^2} \equiv \frac{A}{5-2x} + \frac{B}{5+2x}$. $1 = A(5+2x)+B(5-2x)$. When $x = -\frac{5}{2}$, $1 = 10B \therefore B = \frac{1}{10}$. When $x = \frac{5}{2}$, $1 = 10A \therefore A = \frac{1}{10}$. $\therefore \frac{1}{25-4x^2} \equiv \frac{1}{10(5-2x)} + \frac{1}{10(5+2x)}$.

5)

$$\frac{3x^2-14x+31}{(x-5)(x-3)(x+1)} \equiv \frac{A}{x-5} + \frac{B}{x+1} + \frac{C}{x-3}.$$

$$3x^2-14x+31 = A(x+1)(x-3)+B(x-5)(x-3)+C(x-5)(x+1).$$

When $x=-1$, $3+14+31 = 24B \therefore B = 2$. When $x=3$, $27-42+31 = -8C \therefore C = -2$. When $x=5$, $75-70+31 = 12A \therefore A = 3$. $\therefore \frac{3x^2-14x+31}{(x-5)(x-3)(x+1)} \equiv \frac{3}{x-5} + \frac{2}{x+1} - \frac{2}{x-3}$.

6 a)

$$\frac{4x^2-25x+61}{(x-5)(x-2)(x+1)} \equiv \frac{A}{x-5} + \frac{B}{x-2} + \frac{C}{x+1}.$$

$$4x^2-25x+61 = A(x-2)(x+1)+B(x-5)(x-1)+C(x-5)(x-2).$$

When $x=5$, $100-125+61 = 18A \therefore A = 2$. When $x=2$, $16-50+61 = -9B \therefore B = -3$. When $x=-1$, $4+25+61 = 18C \therefore C = 5$. $\therefore \frac{4x^2-25x+61}{(x-5)(x-2)(x+1)} \equiv \frac{2}{x-5} - \frac{3}{x-2} + \frac{5}{x+1}$.

6 b)

$\frac{2}{(x-4)(x+3)(2x-1)} \equiv \frac{A}{x-4} + \frac{B}{x+3} + \frac{C}{2x-1}$. $2 = A(x+3)(2x-1)+B(x-4)(2x-1)+C(x+3)(x-4)$. When $x=4$, $2 = 49A \therefore A = \frac{2}{49}$. When $x=-3$, $2 = 49B \therefore B = \frac{2}{49}$. When $x = \frac{1}{2}$, $2 = -\frac{49}{4}C \therefore C = -\frac{8}{49}$. $\therefore \frac{2}{(x-4)(x+3)(2x-1)} \equiv \frac{2}{49(x-4)} + \frac{2}{49(x+3)} - \frac{8}{49(2x-1)}$.

7 a)

$f(\frac{1}{2}) = \frac{1}{2} - 2 - \frac{15}{2} + 9 = 0 \therefore (2x-1)$ is a factor.

7 b)

$4x^3 - 8x^2 - 15x + 9 = (2x-1)(2x^2 - 3x - 9) = (2x-1)(2x+3)(x-3)$.

7 c)

$\frac{5}{4x^3 - 8x^2 - 15x + 9} = \frac{5}{(2x-1)(2x+3)(x-3)} \equiv \frac{A}{2x-1} + \frac{B}{2x+3} + \frac{C}{x-3}$. $5 = A(2x+3)(x-3) + B(2x-1)(x-3) + C(2x-1)(2x+3)$. When $x = \frac{1}{2}$, $5 = -10A \therefore A = -\frac{1}{2}$. When $x = -\frac{3}{2}$, $5 = 18B \therefore B = \frac{5}{18}$. When $x = 3$, $5 = 45C \therefore C = \frac{1}{9}$. $\therefore \frac{5}{4x^3 - 8x^2 - 15x + 9} = -\frac{1}{2(2x-1)} + \frac{5}{18(2x+3)} + \frac{1}{9(x-3)}$.

8 a)

Lisa has not multiplied both sides by $(x+2)(x^2-1)$.

8 b)

$\frac{1}{(x+2)(x^2-1)} \equiv \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{x+1}$. $1 = A(x-1)(x+1) + B(x+2)(x+1) + C(x+2)(x-1)$. When $x = -2$, $1 = 3A \therefore A = \frac{1}{3}$. When $x = 1$, $1 = 6B \therefore B = \frac{1}{6}$. When $x = -1$, $1 = -2C \therefore C = -\frac{1}{2}$. $\frac{1}{(x+2)(x^2-1)} \equiv \frac{1}{3(x+2)} + \frac{1}{6(x-1)} - \frac{1}{2(x+1)}$.

9)

$\frac{7x^2 + 21x + 12}{(x+1)^2(x+3)} \equiv \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+3}$. $7x^2 + 21x + 12 = A(x+1)(x+3) + B(x+3) + C(x+1)^2$. When $x = -1$, $-2 = 2B \therefore B = -1$. When $x = -3$, $12 = 4C \therefore C = 3$. When $x = 0$, $12 = 3A + 3B + C \therefore A = 4$. $\frac{7x^2 + 21x + 12}{(x+1)^2(x+3)} \equiv \frac{4}{x+1} - \frac{1}{(x+1)^2} + \frac{3}{x+3}$.

10)

$\frac{32 - 5x - x^2}{(x-2)^2(x+4)} \equiv \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+4}$. $32 - 5x - x^2 = A(x-2)(x+4) + B(x+4) + C(x-2)^2$. When $x = 2$, $18 = 6B \therefore B = 3$. When $x = -4$, $36 = 36C \therefore C = 1$. When $x = 0$, $32 = -8A + 4B + 4C \therefore A = -2$. $\frac{32 - 5x - x^2}{(x-2)^2(x+4)} \equiv -\frac{2}{x-2} + \frac{3}{(x-2)^2} + \frac{1}{x+4}$.

11 a)

$\frac{3}{(x-1)(x+2)^2} \equiv \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$. $3 = A(x+2)^2 + B(x-1)(x+2) + C(x-1)$. When $x = 1$, $3 = 9A \therefore A = \frac{1}{3}$. When $x = -2$, $3 = -3C \therefore C = -1$. When $x = 0$, $3 = 4A - 2B - C \therefore B = -\frac{1}{3}$. $\frac{3}{(x-1)(x+2)^2} \equiv \frac{1}{3(x-1)} - \frac{1}{3(x+2)} - \frac{1}{(x+2)^2}$.

11 b)

$\frac{2x+1}{(2x-1)(x+1)^2} \equiv \frac{A}{2x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$. $2x+1 = A(x+1)^2 + B(2x-1)(x+1) + C(2x-1)$. When $x = -1$, $-1 = -3C \therefore C = \frac{1}{3}$. When $x = \frac{1}{2}$, $2 = \frac{9}{4}A \therefore A = \frac{8}{9}$. When $x = 0$, $1 = A - B - C \therefore B = -\frac{4}{9}$. $\frac{2x+1}{(2x-1)(x+1)^2} \equiv \frac{8}{9(2x-1)} - \frac{4}{9(x+1)} + \frac{1}{3(x+1)^2}$.

11 c)

$\frac{2}{x^2(x-1)} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$. $2 = Ax(x-1) + B(x-1) + Cx^2$. When $x = 0$, $2 = -B \therefore B = -2$. When $x = 1$, $2 = C \therefore C = 2$. When $x = 2$, $2 = 2A + B + 4C \therefore A = -2$. $\frac{2}{x^2(x-1)} \equiv -\frac{2}{x} - \frac{2}{x^2} + \frac{2}{x-1}$.

11 d)

$\frac{x+3}{(x-1)^2} \equiv \frac{A}{x-1} + \frac{B}{(x-1)^2}$. $x+3 = A(x-1)^2 + B(x-1)$. When $x = 0$, $3 = A - B$. When $x = 2$, $5 = A + B$. $\therefore A = 1$, $B = 4$. $\frac{x+3}{(x-1)^2} \equiv \frac{1}{x-1} + \frac{4}{(x-1)^2}$.

12 a)

A.

12 b)

B.

12 c)

$x^2 - x - 6 = (x-3)(x+2)$. A.

12 d)

$x^3 + x^2 - 8x - 12 = (x-3)(x^2+4x+4) = (x-3)(x+2)^2$. B.

12 e)

$x^3 - 4x^2 - 3x + 18 = (x-3)(x^2 - x - 6) = (x-3)(x-3)(x+2)$. C.

13 a)

$\frac{2x+3}{x^2+3x-10} = \frac{2x+3}{(x+5)(x-2)} \equiv \frac{A}{x+5} + \frac{B}{x-2}$. $2x+3 = A(x-2) + B(x+5)$. When $x = 2$, $7 = 7B \therefore B = 1$. When $x = -5$, $-7 = -7A \therefore A = 1$. $\frac{2x+3}{x^2+3x-10} \equiv \frac{1}{x+5} + \frac{1}{x-2}$.

13 b)

$\frac{4}{x^3-4x^2+4x} = \frac{4}{x(x-2)^2} \equiv \frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$. $4 = A(x-2)^2 + Bx(x-2) + Cx$. When $x = 0$, $4 = 4A \therefore A = 1$. When $x = 2$, $4 = 2C \therefore C = 2$. When $x = 1$, $4 = A - B + C \therefore B = -1$. $\frac{4}{x^3-4x^2+4x} \equiv \frac{1}{x} - \frac{1}{x-2} + \frac{2}{(x-2)^2}$.

13 c)

$\dfrac{5x+1}{4x^3+8x^2-x-2} = \dfrac{5x+1}{(x+2)(2x-1)(2x+1)} \equiv \dfrac{A}{x+2} + \dfrac{B}{2x-1} + \dfrac{C}{2x+1}$. $5x+1 = A(2x-1)(2x+1) + B(x+2)(2x+1) + C(x+2)(2x-1)$. When $x = -2$, $-9 = 15A \therefore A = -\frac{9}{15} = -\frac{3}{5}$. When $x = \frac{1}{2}$, $\frac{7}{2} = 5B \therefore B = \frac{7}{10}$. When $x = -\frac{1}{2}$, $-\frac{3}{2} = -3C \therefore C = \frac{1}{2}$. $\dfrac{5x+1}{4x^3+8x^2-x-2} \equiv -\dfrac{3}{5(x+2)} + \dfrac{7}{10(2x-1)} + \dfrac{1}{2(2x+1)}$.

13 d)

$\dfrac{2}{2x^3+x^2-4x-3} = \dfrac{2}{(x+1)^2(2x-3)} \equiv \dfrac{A}{x+1} + \dfrac{B}{(x+1)^2} + \dfrac{C}{2x-3}$. $2 = A(x+1)(2x-3) + B(2x-3) + C(x+1)^2$. When $x = -1$, $2 = -5B \therefore B = -\frac{2}{5}$. When $x = \frac{3}{2}$, $2 = \frac{25}{4}C \therefore C = \frac{8}{25}$. When $x = 0$, $2 = -3A - 3B + C \therefore A = -\frac{4}{25}$. $\dfrac{2}{2x^3+x^2-4x-3} \equiv -\dfrac{4}{25(x+1)} - \dfrac{2}{5(x+1)^2} + \dfrac{8}{25(2x-3)}$.

14 a)

$\dfrac{5x-1}{(x-2)(x+7)} \equiv \dfrac{A}{x-2} + \dfrac{B}{x+7}$. $5x-1 = A(x+7) + B(x-2)$. When $x = -7$, $-36 = -9B \therefore B = 4$. When $x = 2$, $9 = 9A \therefore A = 1$. $\dfrac{5x-1}{(x-2)(x+7)} \equiv \dfrac{1}{x-2} + \dfrac{4}{x+7}$.

14 b)

$\dfrac{3x^2-19x+44}{(x+7)(x-2)^2} \equiv \dfrac{A}{x+7} + \dfrac{B}{x-2} + \dfrac{C}{(x-2)^2}$. $5x^2-9x+16 = A(x-2)^2 + B(x+7)(x-2) + C(x+7)$. When $x = 2$, $18 = 9C \therefore C = 2$. When $x = -7$, $324 = 81A \therefore A = 4$. When $x = 0$, $44 = 4A - 14B + 7C \therefore B = -1$. $\dfrac{3x^2-19x+44}{(x+7)(x-2)^2} \equiv \dfrac{4}{x+7} - \dfrac{1}{x-2} + \dfrac{2}{(x-2)^2}$.

14 c)

$$\begin{aligned}
\frac{1}{x-2} + \frac{4}{x+7} &= \frac{4}{x+7} - \frac{1}{x-2} + \frac{2}{(x-2)^2}, \\
\frac{1}{x-2} &= -\frac{1}{x-2} + \frac{2}{(x-2)^2}, \\
\frac{2}{x-2} = \frac{2}{(x-2)^2} &\Rightarrow 2(x-2)^2 = 2(x-2), \\
x^2 - 4x + 4 &= x - 2, \\
x^2 - 5x + 6 &= 0 \therefore x = 2,\ x = 3.
\end{aligned}$$

$\therefore x = 3$.

15)

$$\begin{aligned}
5x + 19 &= (x+p)^2 - (x+3)(x+p) + 3(x+3) \\
&= x^2 + 2px + p^2 - x^2 - 3x - px - 3p + 3x + 9 \\
&= px + p^2 - 3p + 9.
\end{aligned}$$

$5x + 19 = px + (p^2 - 3p + 9) \therefore p = 5$.

16 a)

$\frac{x+2}{(x-4)(2x+1)} \equiv \frac{A}{x-4}+\frac{B}{2x+1}$. $x+2 = A(2x+1)+B(x-4)$. When $x = -\frac{1}{2}$, $\frac{3}{2} = -\frac{9}{2}B \therefore B = -\frac{1}{3}$. When $x = 4$, $6 = 9A \therefore A = \frac{2}{3}$. $\frac{x+2}{(x-4)(2x+1)} \equiv \frac{2}{3(x-4)} - \frac{1}{3(2x+1)}$.

16 b)

$f(x) = \frac{2}{3}(x-4)^{-1} - \frac{1}{3}(2x+1)^{-1}$. $f'(x) = -\frac{2}{3}(x-4)^{-2} + \frac{2}{3}(2x+1)^{-2}$. Stat point when $f'(x) = 0$.

$$-\frac{2}{3(x-4)^2} + \frac{2}{3(2x+1)^2} = 0,$$
$$\frac{2}{3(2x+1)^2} = \frac{2}{3(x-4)^2} \Rightarrow 6(x-4)^2 = 6(2x+1)^2,$$
$$x^2 - 8x + 16 = 4x^2 + 4x + 1,$$
$$3x^2 + 12x - 15 = 0 \therefore x = 1 \text{ or } x = -5.$$

$(1, -\frac{1}{3})$ or $(-5, -\frac{1}{27})$.

17 a)

$\frac{3x}{(x-2)^2(5-x)} \equiv \frac{A}{x-2}+\frac{B}{(x-2)^2}+\frac{C}{5-x}$. $3x = A(x-2)(5-x)+B(5-x)+C(x-2)^2$. When $x = 2$, $6 = 3B \therefore B = 2$. When $x = 5$, $15 = 9C \therefore C = \frac{5}{3}$. When $x = 0$, $0 = -10A + 5B + 4C \therefore A = \frac{5}{3}$. $\frac{3x}{(x-2)^2(5-x)} \equiv \frac{5}{3(x-2)} + \frac{2}{(x-2)^2} + \frac{5}{3(5-x)}$.

17 b)

$f(x) = \frac{5}{3(x-2)} + \frac{2}{(x-2)^2} + \frac{5}{3(5-x)}$. $f(x) = \frac{5}{3}(x-2)^{-1} + 2(x-2)^{-2} + \frac{5}{3}(5-x)^{-1}$. $f(3) = \frac{9}{2}$. $f'(x) = -\frac{5}{3}(x-2)^{-2} - 4(x-2)^{-3} + \frac{5}{3}(5-x)^{-2}$. $f'(3) = -\frac{21}{4}$. $y - \frac{9}{2} = -\frac{21}{4}(x-3) \Rightarrow 4y - 18 = -21x + 63 \Rightarrow 4y + 21x = 81$ as req.

18 a)

$\frac{2x+1}{(x+3)(2-x)} \equiv \frac{A}{x+3} + \frac{B}{2-x}$. $2x+1 = A(2-x) + B(x+3)$. When $x = 2$, $5 = 5B \therefore B = 1$. When $x = -3$, $-5 = 5A \therefore A = -1$. $\frac{2x+1}{(x+3)(2-x)} \equiv -\frac{1}{x+3} + \frac{1}{2-x} = -\frac{1}{x+3} - \frac{1}{x-2}$.

18 b)

$f(x) = -\frac{1}{x+3} + \frac{1}{2-x} = -(x+3)^{-1} + (2-x)^{-1}$. Increasing when $f'(x) > 0$. $f'(x) = (x+3)^{-2} + (2-x)^{-2} = \frac{1}{(x+3)^2} + \frac{1}{(2-x)^2}$. Since $(x+3)^2$ and $(2-x)^2$ are always positive, the two terms are always positive, $\therefore f(x)$ is always increasing.

19 a)

$\frac{5}{(x+1)(x+2)} \equiv \frac{A}{x+1} + \frac{B}{x+2}$. $5 = A(x+2) + B(x+1)$. When $x = -2$, $5 = -B \therefore B = -5$. When $x = -1$, $5 = A \therefore A = 5$. $\frac{5}{(x+1)(x+2)} \equiv \frac{5}{x+1} - \frac{5}{x+2}$.

19 b)

$f(x) = 5(x+1)^{-1} - 5(x+2)^{-1}$. Increasing when $f'(x) < 0$. $f'(x) = -5(x+1)^{-2} + 5(x+2)^{-2} = -\dfrac{5}{(x+1)^2} + \dfrac{5}{(x+2)^2}$. Since $(x+1)^2$ and $(x+2)^2$ are always positive, and $\dfrac{5}{(x+2)^2} < \dfrac{5}{(x+1)^2}$, $f'(x) < 0$, then $f(x)$ is always decreasing.

20 a)

$\dfrac{1}{(1-2x)(1+x)} \equiv \dfrac{A}{1-2x} + \dfrac{B}{1+x}$. $1 = A(1+x) + B(1-2x)$. When $x = -1$, $1 = 3B \therefore B = \frac{1}{3}$. When $x = \frac{1}{2}$, $1 = \frac{3}{2}A \therefore A = \frac{2}{3}$. $\dfrac{1}{(1-2x)(1+x)} \equiv \dfrac{2}{3(1-2x)} + \dfrac{1}{3(1+x)}$.

20 b)

$$\begin{aligned}
\frac{2}{3(1-2x)} &= \frac{2}{3}(1-2x)^{-1} \\
&= \frac{2}{3}\left[1 + (-1)(-2x) + \frac{(-1)(-2)}{2}(-2x)^2 + \frac{(-1)(-2)(-3)}{6}(-2x)^3 + \cdots\right] \\
&= \frac{2}{3} + \frac{4}{3}x + \frac{8}{3}x^2 + \frac{16}{3}x^3 + \cdots, \\
\frac{1}{3(1+x)} &= \frac{1}{3}(1+x)^{-1} \\
&= \frac{1}{3}\left[1 + (-1)x + \frac{(-1)(-2)}{2}(-x)^2 + \frac{(-1)(-2)(-3)}{6}(-x)^3 + \cdots\right] \\
&= \frac{1}{3} - \frac{1}{3}x + \frac{1}{3}x^2 - \frac{1}{3}x^3 + \cdots, \\
\frac{1}{(1-2x)(1+x)} &= 1 + x + 3x^2 + 5x^3 + \cdots.
\end{aligned}$$

20 c)

$|2x| < 1 \Rightarrow |x| < \frac{1}{2}$ and $|x| < 1 \therefore |x| < \frac{1}{2}$.

21 a)

$\dfrac{1}{(x+1)(x-2)} \equiv \dfrac{A}{x+1} + \dfrac{B}{x-2}$. $1 = A(x-2) + B(x+1)$. When $x = 2$, $1 = 3B \therefore B = \frac{1}{3}$. When $x = -1$, $1 = -3A \therefore A = -\frac{1}{3}$. $\dfrac{1}{(x+1)(x-2)} \equiv -\dfrac{1}{3(x+1)} + \dfrac{1}{3(x-2)}$.

21 b)

$$\begin{aligned}
-\frac{1}{3(x+1)} &= -\frac{1}{3}(1+x)^{-1} \\
&= -\frac{1}{3}\left[1 + (-1)x + \frac{(-1)(-2)}{2}x^2 + \frac{(-1)(-2)(-3)}{6}x^2 + \cdots\right] \\
&= -\frac{1}{3} + \frac{1}{3}x - \frac{1}{3}x^2 + \frac{1}{3}x^3 + \cdots, \\
\frac{1}{3(x-2)} &= \frac{1}{3}(-2+x)^{-1} = \frac{1}{3}\left(-2(1 - \tfrac{1}{2}x)\right)^{-1}
\end{aligned}$$

$$
\begin{aligned}
&= -\frac{1}{6}(1-\tfrac{1}{2}x)^{-1} \\
&= -\frac{1}{6}\left[1+(-1)(-\tfrac{1}{2}x)+\frac{(-1)(-2)}{2}(-\tfrac{1}{2}x)^2+\frac{(-1)(-2)(-3)}{6}(-\tfrac{1}{2}x)^3+\cdots\right] \\
&= -\frac{1}{6}-\frac{1}{12}x-\frac{1}{24}x^2-\frac{1}{48}x^3+\cdots, \\
\frac{1}{(x+1)(x-2)} &= -\frac{1}{2}+\frac{1}{4}x-\frac{3}{8}x^2+\frac{5}{16}x^3+\cdots.
\end{aligned}
$$

21 c)

$|x|<1$ and $|\frac{1}{2}x|<1 \Rightarrow |x|<2 \therefore |x|<1$.

22 a)

$\frac{3x^2-5x+6}{(1-x)^2(3+x)} \equiv \frac{A}{1-x}+\frac{B}{(1-x)^2}+\frac{C}{3+x}$. $3x^2-5x+6 = A(1-x)(3+x)+B(3+x)+C(1-x)^2$. When $x=1$, $4=4B \therefore B=1$. When $x=-3$, $48=16C \therefore C=3$. When $x=0$, $6=3A+3B+C \therefore A=0$. $\frac{3x^2-5x+6}{(1-x)^2(3+x)} \equiv \frac{1}{(1-x)^2}+\frac{3}{3+x}$.

22 b)

$$
\begin{aligned}
\frac{1}{(1-x)^2} &= (1-x)^{-2} \\
&= 1+(-2)(-x)+\frac{(-2)(-3)}{2}(-x)^2+\frac{(-2)(-3)(-4)}{6}(-x)^3+\cdots \\
&= 1+2x+3x^2+4x^3+\cdots, \\
\frac{3}{3+x} &= 3(3+x)^{-1} = 3\left(3(1+\tfrac{1}{3}x)\right)^{-1} = (1+\tfrac{1}{3}x)^{-1} \\
&= 1+(-1)(\tfrac{1}{3}x)+\frac{(-1)(-2)}{2}(\tfrac{1}{3}x)^2+\frac{(-1)(-2)(-3)}{6}(\tfrac{1}{3}x)^3+\cdots \\
&= 1-\frac{1}{3}x+\frac{1}{9}x^2-\frac{1}{27}x^3+\cdots, \\
\frac{3x^2-5x+6}{(1-x)^2(3+x)} &= 2+\frac{5}{3}x+\frac{28}{9}x^2+\frac{107}{27}x^3+\cdots.
\end{aligned}
$$

22 c)

$|x|<1$ and $|\frac{1}{3}x|<1 \Rightarrow |x|<3 \therefore |x|<1$.

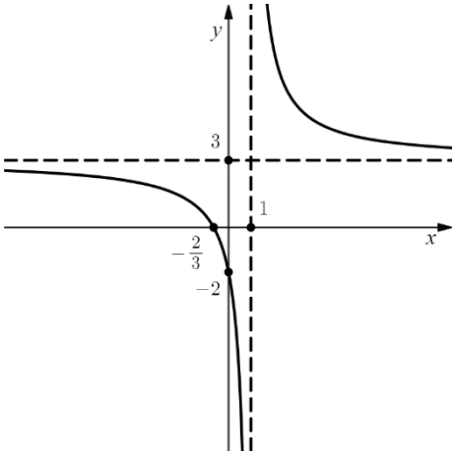
23)

$8x^2+4x+1 = A(x^2+1)+(Bx+C)(2x-1)$. When $x=\frac{1}{2}$, $5=\frac{5}{4}A \therefore A=4$. When $x=0$, $1=A-C \therefore C=3$. When $x=1$, $13=2A+B+C \therefore B=2$.

24 a)

$3x+2 = A(x-1)+B$. When $x=1$, $B=5$. When $x=0$, $2=-A+B \therefore A=3$.

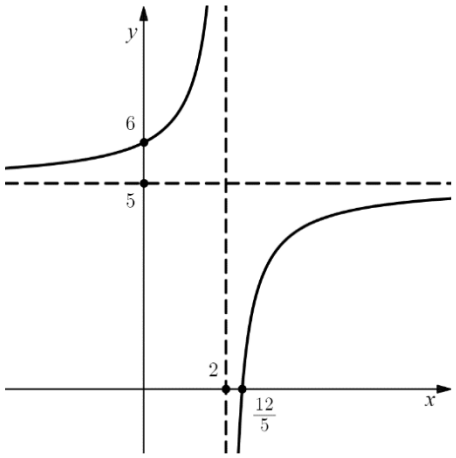
24 b)



25 a)

$12 - 5x = A(2 - x) - B$. When $x = 2$, $2 = -B \therefore B = -2$. When $x = 0$, $12 = 2A - B \therefore A = 5$.

25 b)



26 a)

Compare x^2 terms: $-4 = -4A \therefore A = 1$. Compare x terms: $-7 = A - 4B \therefore B = 2$. Compare constant terms: $10 = B + C \therefore C = 8$.

26 b)

Compare x^2 terms: $2 = -A \therefore A = -2$. Compare x terms: $0 = 3A - B \therefore B = -6$. Compare constant terms: $15 = 3B + C \therefore C = 33$.

26 c)

Compare x^3 terms: $3 = A \therefore A = 3$. Compare x^2 terms: $2 \therefore B = 2$. Compare x terms: $0 = A + C \therefore C = -3$. Compare constant terms: $0 = B + D \therefore D = -2$.

26 d)

Compare x^3 terms: $1 = A \therefore A = 1$. Compare x^2 terms: $0 = A + B \therefore B = -1$. Compare constant terms: $2 = C \therefore C = 2$. Compare x terms: $0 = B + C + D \therefore D = -1$.

27)

$x^2 - 3x - 13 = A(x-5) + Bx(x-5) + C$. When $x = 5$, $-3 = C \therefore C = -3$. When $x = 0$, $-13 = -5A + C \therefore A = 2$. When $x = 1$, $-15 = -4A - 4B + C \therefore B = 1$.

28)

$2+5x-5x^2 = Ax(1-x)+B$. When $x = 0$, $2 = B \therefore B = 2$. When $x = -1$, $-8 = -2A+2 \therefore A = 5$.

29)

$\frac{x^3}{x^2+3x-4} = \frac{x^3}{(x-1)(x+4)}$. $x^3 = A(x+4)(x-1) + Bx(x+4)(x-1) + C(x+4) + D(x-1)$. When $x = -4$, $-64 = -5D \therefore D = \frac{64}{5}$. When $x = 1$, $1 = 5C \therefore C = \frac{1}{5}$. When $x = 0$, $0 = -4A + 4C - D \therefore A = -3$. When $x = 2$, $8 = 6A + 12B + 6C + D \therefore B = 1$.

30)

$$x^4 - 1 = Ax(x-2)(x-1)(x+4) + B(x-2)(x-1)(x+4) + C(x+4)(x-1) + D(x-2)(x-1).$$

When $x = 2$, $15 = 6C \therefore C = \frac{5}{2}$. When $x = -4$, $255 = 30D \therefore D = \frac{17}{2}$. When $x = 0$, $-1 = 8B - 4C + 2D \therefore B = -1$. When $x = -1$, $0 = -18A + 18B - 6C + 6D \therefore A = 1$.

31)

$2x^4 - x^3 - x + 4 = Ax^2(x^2+1) + Bx(x^2+1) + C(x^2+1) + D$. When $x = 0$, $4 = C + D$. Compare x^4 terms: $2 = A \therefore A = 2$. Compare x^3 terms: $-1 = B \therefore B = -1$. When $x = 1$, $4 = 2A+2B+2C+D$. $4 = 4 - 2 + 2C + D \Rightarrow 2 = 2C + D$. $C = -2$, $D = 6$.

32 a)

$\frac{4p+5}{(p+1)(p+2)} \equiv \frac{A}{p+1} + \frac{B}{p+2}$. $4p + 5 = A(p+2) + B(p+1)$. When $p = -2$, $-3 = -B \therefore B = 3$. When $p = -1$, $1 = A \therefore A = 1$. $\therefore \frac{4p+5}{(p+1)(p+2)} \equiv \frac{1}{p+1} + \frac{3}{p+2}$.

32 b)

Term 2− term 1:

$$\frac{3}{p+2} - \frac{1}{p+1} = \frac{3(p+1)}{(p+1)(p+2)} - \frac{p+2}{(p+1)(p+2)}, \qquad d = \frac{2p+1}{(p+1)(p+2)}.$$

Term 2+ difference:

$$\frac{2p+1}{(p+1)(p+2)} + \frac{3}{p+2} = \frac{2p+1}{(p+1)(p+2)} + \frac{3(p+1)}{(p+1)(p+2)} = \frac{5p+4}{(p+1)(p+2)}.$$

32 c)

$$\frac{4p+5}{(p+1)(p+2)} + \frac{5p+4}{(p+1)(p+2)} = 1,$$

$$\frac{9p+9}{(p+1)(p+2)} = 1,$$

$$9p+9 = (p+1)(p+2) \Rightarrow 9p+9 = p^2+3p+2,$$

$$p^2 - 6p - 7 = 0 \therefore p = 7 \text{ or } p = -1.$$

$p \neq -1$ as not valid. $u_1 = \frac{1}{8}$, $u_2 = \frac{1}{3}$, $u_3 = \frac{13}{24}$.